Probability Basics

Question 1: A die is rolled. What is the probability of getting: (a) An even number (b) A number greater than 4

(a) p(An even number) = 3 / 6 = ½
(b) p(A number greater than 4)= 2/6 = 1/3

Question 2: In a class of 50 students: 20 like Mathematics (M) 15 like Science (S) 5 like both subjects What is the probability that a student chosen at random likes Mathematics or Science?

n(M) = 20
n(S) = 15
n(M & S) = 5
n(M or S) = n(M) + n(S) - n(M & S) = 20 + 15 - 5 = 30
p(M or S) = 30 / 50 = 0.6
Hence, the probability that a randomly chosen student likes Mathematics or Science is 0.6

Question 3: A bag has 3 red and 2 blue balls. If one ball is drawn randomly and is red, what is the probability that the next ball is also red (without replacement)?

p(red) = 3 / 5
p(second ball is also red without replacement) = 2 / 4 = 0.5

Question 4: The population of a school is divided into 60% boys and 40% girls. If you want equal representation of both genders in the sample, which method should you use: Simple Random Sampling or Stratified Sampling? Why?

We will use **Stratified Sampling** because the population is divided into groups (strata) based on a characteristic—in this case, gender (boys and girls). Then, samples are selected separately from each group.This method ensures that both boys and girls are represented in the sample according to the desired proportion. If you want **equal representation**, you can deliberately select the same number of boys and girls.

With **Simple Random Sampling**, every student has an equal chance of being selected, but there is no guarantee that boys and girls will be equally represented. The sample might end up with more boys or more girls just by chance.

Question 5: The average height of 1000 students = 160 cm. A sample of 100 students shows an average height = 158 cm. Find the sampling error.

Given:

- Population mean height (μ) = 160 cm
- Sample mean height (x bar) = 158 cm

Formula for Sampling Error
Sampling Error = Sample Mean−Population Mean = 158 - 160 = -2 cm

The negative sign indicates that the sample underestimates the population mean by 2 cm. Therefore, the sampling error is −2cm (or 2 cm in magnitude).

Question 6: The population mean salary is ₹50,000 with σ = ₹5,000. If we take a sample of 100 employees, what is the standard error of the mean (SEM)?
Given:
Population mean salary = ₹50,000
Population standard deviation (σ) = ₹5,000
Sample size (n) = 100

Formula for Standard Error of the Mean (SEM)

SEM = σ / sqrt n

SEM = 5000 / root 100 = 500

The standard error of ₹500 indicates the typical amount by which the sample mean salary is expected to vary from the true population mean when samples of 100 employees are repeatedly drawn.

Question 7: In a group of 100 students: 40 like Cricket (C) 30 like Football (F) 10 like both Cricket and Football Find the probability that a student likes at least one sport.
Given:

- Total students = 100
- Like Cricket (C) = 40
- Like Football (F) = 30
- Like both Cricket and Football (C∩F) = 10

n(C∪F)=n(C)+n(F)−n(C∩F) =40+30−10= 60
P(a student likes at least one sport) = 60 / 100 = 0.6

Question 8: From a deck of 52 cards, two cards are drawn without replacement. What is the probability that both are Aces?

P (one card to be Ace) = 4 / 52 = 1 / 13
p(another card also Ace without 1st replaced) = 3 / 51
P (both Aces) = 1/ 13 * 1/ 17 = 1 / 211 = 0.00452
Hence, the probability that both cards drawn are Aces is 1 / 221 = 0.452 %

Question 9: A factory produces bulbs with 2% defective rate. If 5 bulbs are chosen at random, what is the probability that all are non-defective?

Defective rate = 2% = 0.02
Non Defective = 1- 0.02 = 0.98
P(5 Non Defective balls) = (0.98) ^ 5 = 0.9039 = 90.39 %

Question 10: Differentiate between discrete and continuous random variables with examples.

Discrete Random Variable: Takes countable, distinct values obtained by counting

Examples

- Number of cars passing through a toll booth in an hour.
- Number of emails received in a day.
- Number of sixes obtained when rolling a die 10 times.

✅ **Continuous Random Variable:** Takes any value within a range and is obtained by measurement

Examples

- Height of students in a class.
- Time taken to complete a race.
- Amount of rainfall in a day.